## EPISODE RECAP

$P_{X,Y}$ is the joint probability mass function of $X, Y$ if

$$P_{X,Y}(x,y) = P(X=x, Y=y)$$

Note that $P_{X,Y}(x,y) \geq 0$ and $\sum_x \sum_y P_{X,Y}(x,y) = 1$

$P_{X,Y}$ is the joint probability density function of $X, Y$ if

$$P(X \in A, Y \in B) = \int_A \int_B P_{X,Y}(x,y)\,dy\,dx$$

Note that $P_{X,Y}(x,y) \geq 0$ and $\iint P_{X,Y}(x,y)\,dx\,dy = 1$

↳ integrate over the whole domain

The marginals $P_X, P_Y$ are defined:

| DISCRETE CASE | CONTINUOUS CASE |
|---|---|
| $P_X(x) = \sum_y P_{X,Y}(x,y) = P(X=x)$ | $P_X(x) = \int P_{X,Y}(x,y)\,dy$ |
| $P_Y(y) = \sum_x P_{X,Y}(x,y) = P(Y=y)$ | $P_Y(y) = \int P_{X,Y}(x,y)\,dx$ |

$X, Y$ are INDEPENDENT if and only if

$$P_{X,Y}(x,y) = P_X(x)\,P_Y(y) \quad \forall x,y$$

or equivalently if

$$P(X \leq x, Y \leq y) = P(X \leq x)\,P(Y \leq y) \quad \forall x,y$$

The conditional density $P_{Y|X}$ of $Y$ given $X$ is defined:

$$P_{Y|X}(y|x) = \frac{P_{X,Y}(x,y)}{P_X(x)}$$

Note that $P_X(x) > 0$. The same definition holds for a conditional probability mass function $P_{Y|X}(y|x)$

Consequently we have the following factorisation for joint pdf and pmf:

$$P_{X,Y}(x,y) = P_X(x)\, P_{Y|X}(y|x)$$

BAYES THEOREM: $$P_{X|Y}(x|y) = \frac{P_X(x)\, P_{Y|X}(y|x)}{P_Y(y)}$$

note that this denominator is often calculated by marginalising the joint $P_{X,Y}$

$$P_Y(y) = \sum_x P_{X,Y}(x,y)$$

or

$$P_Y(y) = \int P_{X,Y}(x,y)\, dx$$